ESE 2113 / MODULE 02

Graphs, Percentiles and Data Interpretation

Detailed lecture and self-study edition

Second Year, First Semester • English • Excel + R

All numerical teaching datasets are illustrative unless explicitly stated otherwise.

A graph answers a specific question

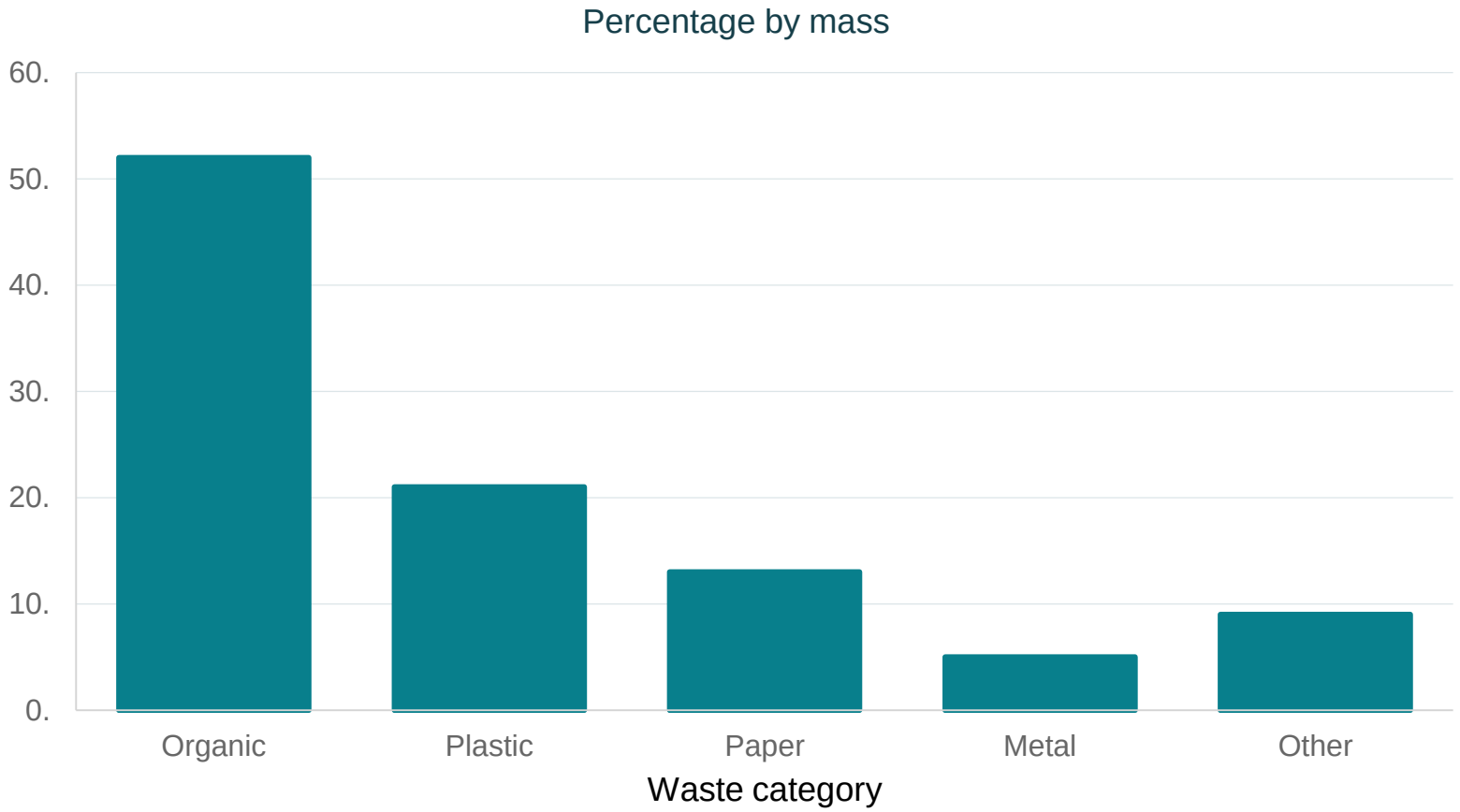
A bar chart compares categories. A histogram describes the distribution of a quantitative variable.

A time series shows temporal order. A scatterplot examines a relationship between paired numerical variables.

A boxplot compares location and spread across groups, but hides multimodality.

Select a graph using the question and variable types. A decorative 3-D view makes accurate comparisons harder.

Categorical comparison: waste composition



How to read this graph

Illustrative waste audit, total = 100%.

Bars are separate because the classes are categories.

Organic waste accounts for 52% of mass, not necessarily 52% of items.

Illustrative data or theoretical distribution. Exact plotted values remain editable.

Bar chart versus histogram

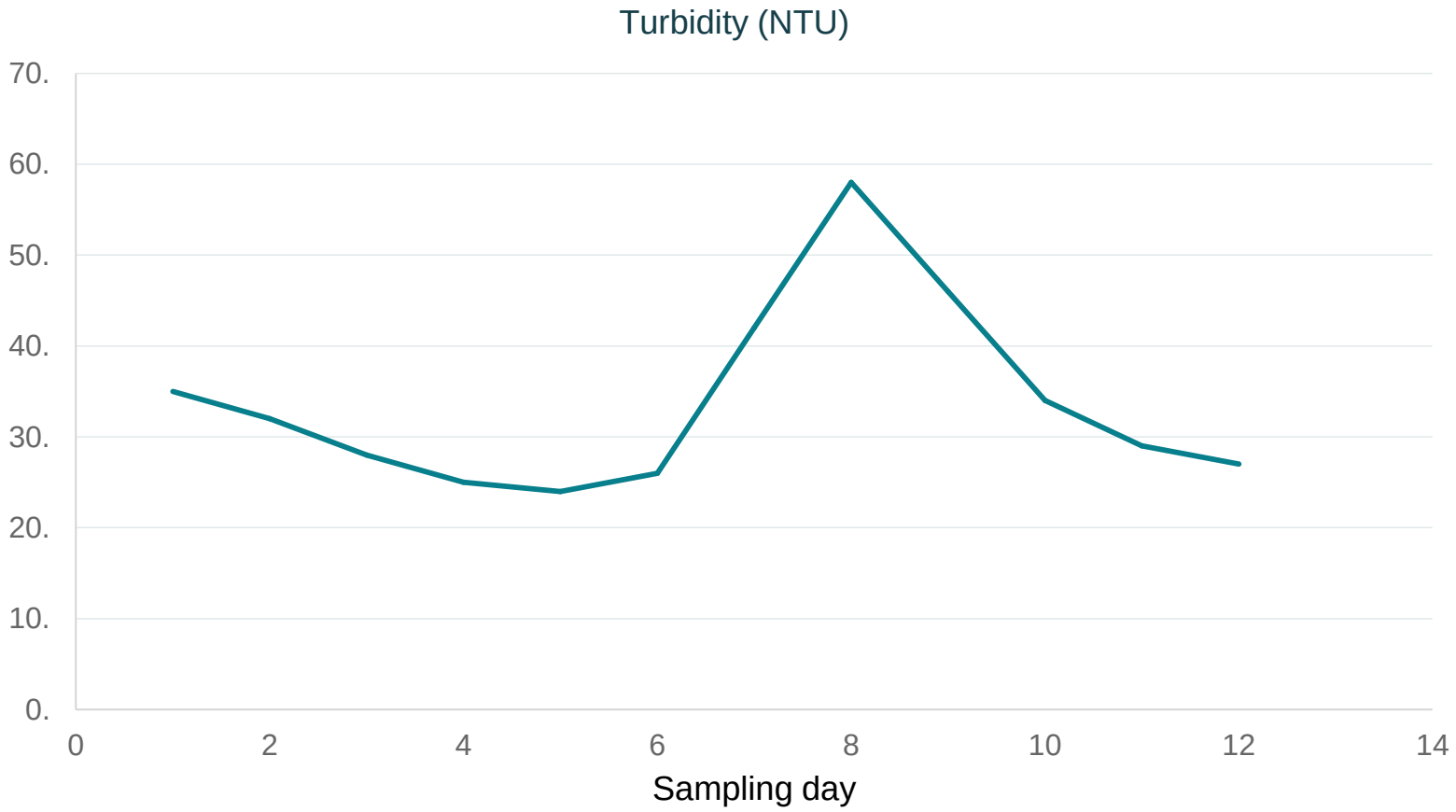
In a bar chart, category order can be chosen to clarify comparison. In a histogram, numerical order is fixed.

For equal-width histogram bins, frequency is proportional to area and height. For unequal bins, use density.

A bar-chart baseline should normally start at zero because viewers compare bar lengths.

A line-graph axis may start elsewhere, but the range must remain clear and should not exaggerate tiny changes.

Time series: preserving the sequence



How to read this graph

The peak occurs on day 8, after an illustrative rainfall event.

A histogram would hide when the high values occurred.

Temporal coincidence alone does not establish the cause.

Illustrative data or theoretical distribution. Exact plotted values remain editable.

Percentiles and quantile conventions

The pth percentile is a value with approximately p% of observations at or below it. Quantiles generalize percentiles to fractions.

Sort data before locating a sample quantile. Interpolation rules differ among software packages, particularly for small n.

In this course the default is R type=7, matching Excel PERCENTILE.INC: $h = 1 + (n-1)p$.

For x=(2,4,6,8,10), Q1 at h=2 is 4 and Q3 at h=4 is 8. Always state the method when comparing software.

Five-number summary and outlier fences

Quantity	Value	Interpretation
Minimum	2	Smallest observation
Q1	4	25th percentile
Median	6	50th percentile
Q3	8	75th percentile
Maximum	10	Largest observation
IQR / fences	4 / −2 to 14	Q1−1.5IQR and Q3+1.5IQR

Example data: 2,4,6,8,10 mg/L. Tukey boxplot hinges can differ slightly from type-7 quartiles.

Reading a boxplot correctly

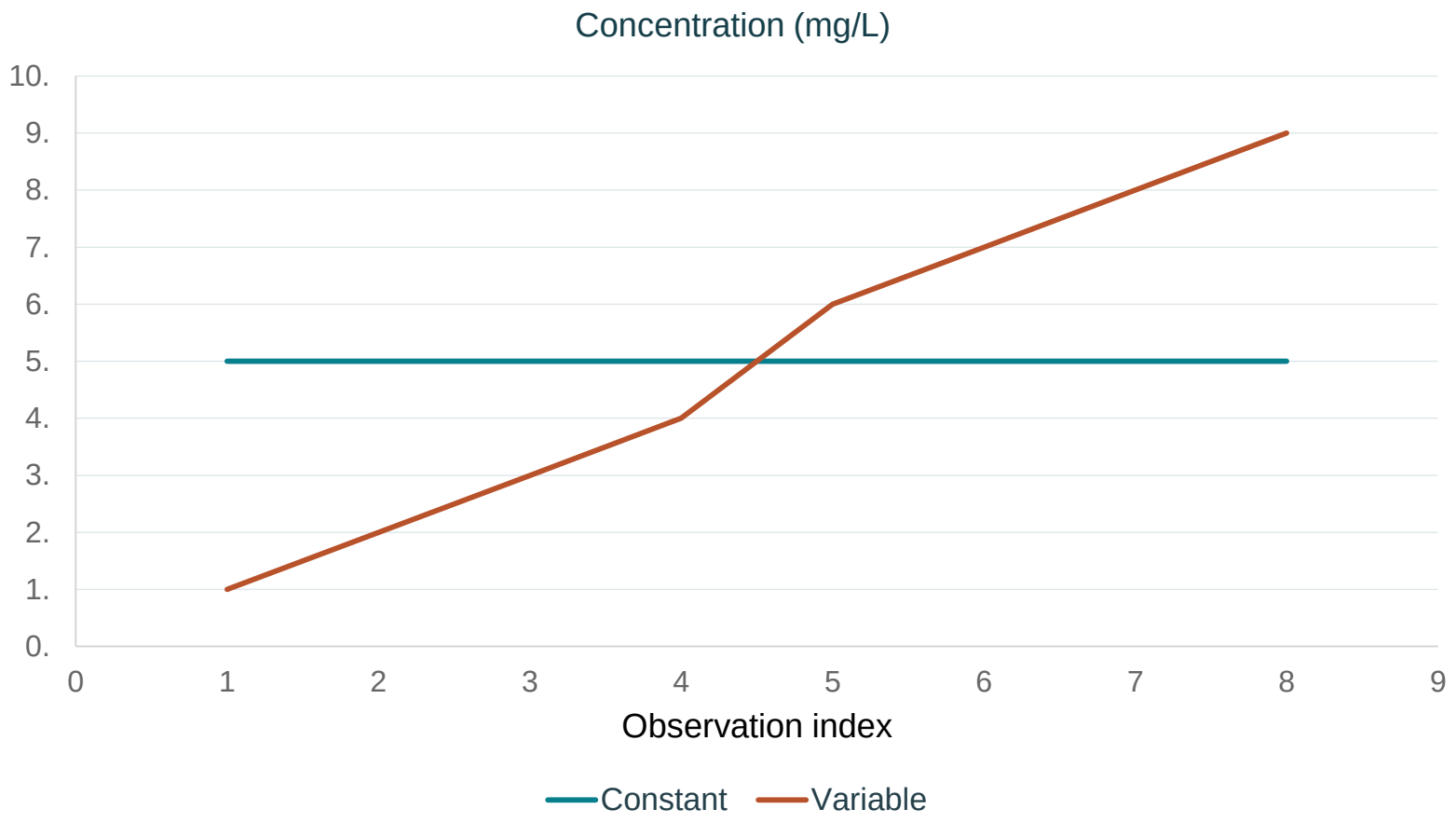
The box spans Q1 to Q3, so it represents the middle half of the sample. The internal line marks the median.

In a Tukey-style boxplot, whiskers reach the most extreme observations within the 1.5 IQR fences. They need not reach the minimum and maximum.

Points beyond the whiskers are flags for investigation. They are not automatically errors.

Unequal box halves or whiskers suggest asymmetry. They do not establish a probability distribution.

Same mean, different patterns



How to read this graph

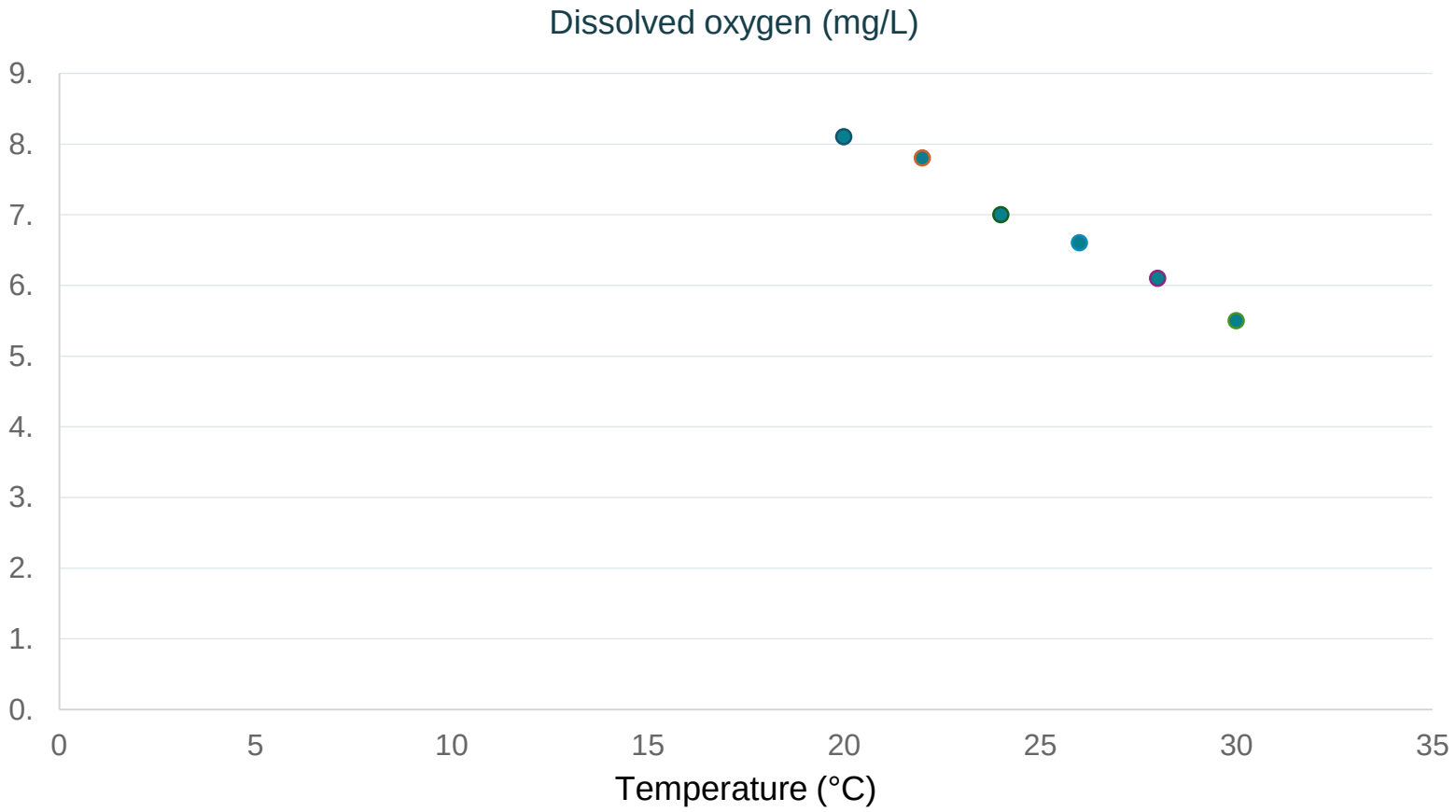
Both series have mean 5 mg/L.

One has no variation, while the other ranges from 1 to 9.

Always pair a centre estimate with a spread measure.

Illustrative data or theoretical distribution. Exact plotted values remain editable.

Scatterplot: paired temperature and oxygen



How to read this graph

Each point is one water sample with two measurements.

The association is negative and approximately linear.

The order of rows does not matter, but correct pairing does.

Illustrative data or theoretical distribution. Exact plotted values remain editable.

Frequency polygon and empirical CDF

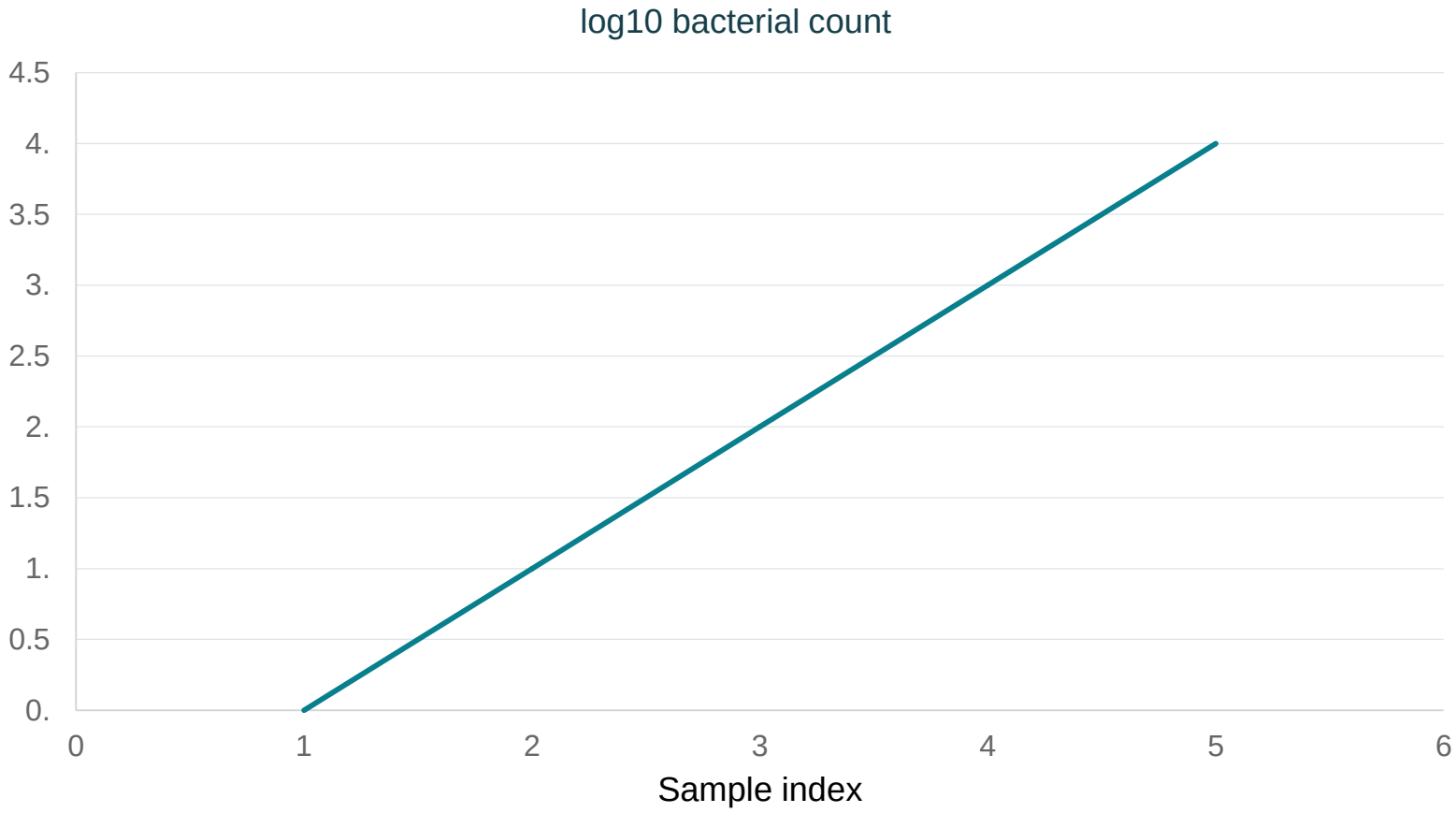
A frequency polygon connects frequencies at class midpoints. With several samples, common boundaries make comparisons clearer.

The empirical CDF is F□(x)=number of observations ≤x divided by n. It is a step function with jumps at observations.

An ogive interpolates grouped cumulative values. It is not identical to the raw-data empirical CDF.

CDFs directly answer “what fraction is at most x?” without a bin-width choice.

Log scale for orders of magnitude



How to read this graph

Original counts: 1, 10, 100, 1000, 10000.

Equal vertical steps now represent tenfold ratios.

Zero and negative values have no real logarithm. Do not add arbitrary constants without justification.

Illustrative data or theoretical distribution. Exact plotted values remain editable.

Detection limits and missing values

A result reported as <0.05 mg/L is left-censored. Its true value is somewhere below that limit, not known to be zero.

Record both the reported limit and censoring status. A graph can display censored observations using a distinct symbol.

Replacing every nondetect with zero or half the limit can bias means and relationships. The effect depends on censoring extent and distribution.

This course flags censoring as a limitation. Specialist methods such as censored regression require further study.

Uncertainty bars and honest presentation

SD bars summarize variability of individual measurements. SE bars summarize precision of the estimated mean. A confidence interval targets uncertainty in a parameter.

These are not interchangeable. Label the bar type and sample size explicitly.

Do not infer a formal test solely from whether error bars overlap. The comparison depends on the intervals and design.

Show units, group definitions and data provenance. Avoid dual axes when they could manufacture apparent agreement.

Graph diagnosis | Try it

An analyst compares mean nitrate across sites using bars with a y-axis beginning at 4.8 mg/L. Means are 5.0 and 5.2. What can mislead?

For 1,2,3,4,20, calculate type-7 Q1, median, Q3 and upper fence.

Graph diagnosis | Solution

The truncated bar axis exaggerates the 0.2 mg/L difference. Use a zero baseline or a clearly labelled point-and-interval plot.

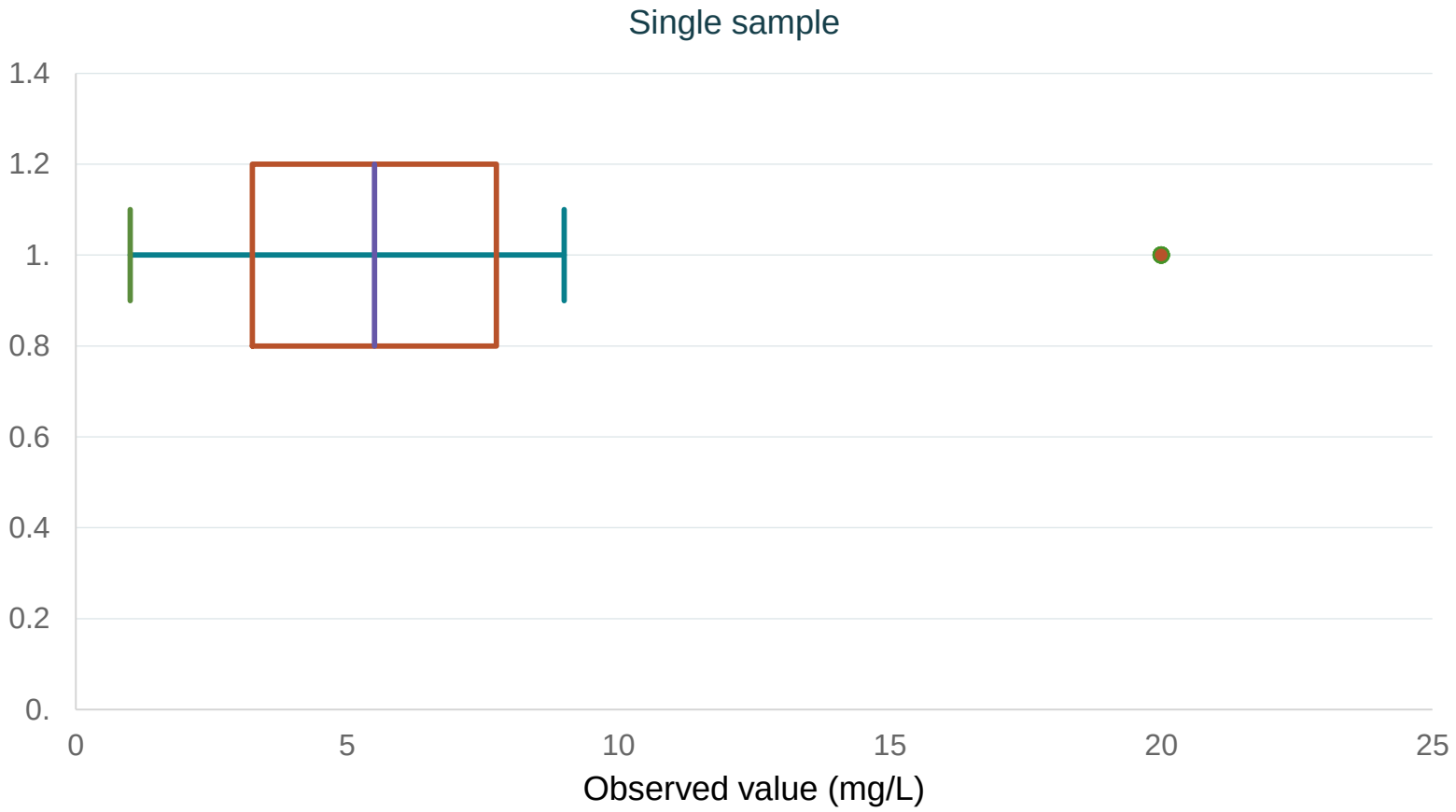
Q1=2, median=3, Q3=4, IQR=2 and upper fence=7. The value 20 is flagged, not automatically deleted.

From ordered values to a boxplot

Step	Computation
Data	1,2,3,4,5,6,7,8,9,20
Type-7 quartiles	Q1=3.25; median=5.5; Q3=7.75
IQR	7.75−3.25=4.5
Fences	−3.5 and 14.5
Whisker endpoints	1 and 9
Flagged observation	20

This explicitly defined boxplot uses type-7 quartiles. Other boxplot algorithms may choose different hinges.

Boxplot: quartiles, whiskers and one outlier



How to read this graph

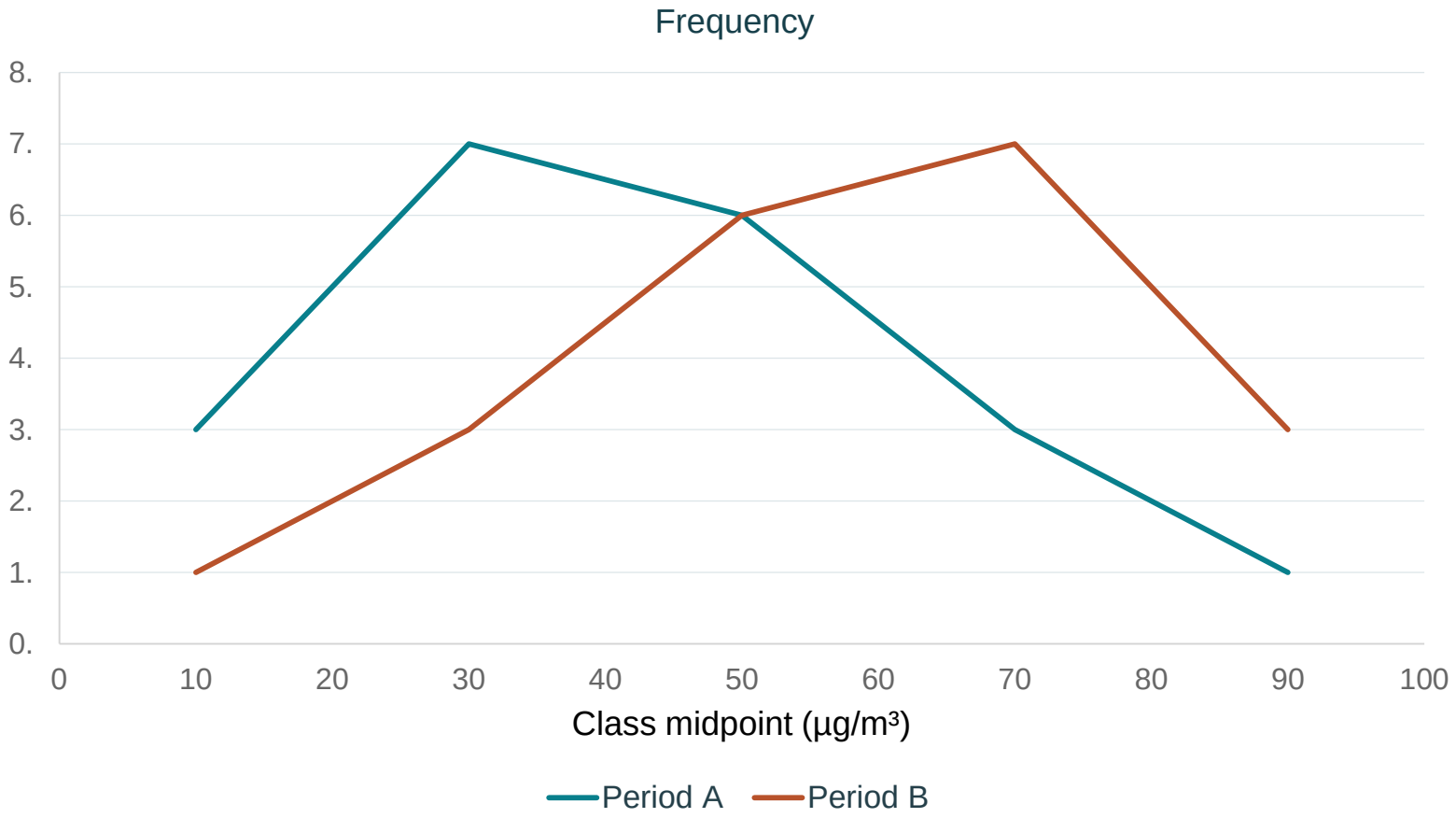
The box spans Q1=3.25 to Q3=7.75. Median=5.5.

Whiskers end at 1 and 9. The value 20 is beyond the upper fence 14.5.

This editable chart uses the type-7 quartiles defined in the calculation table.

Illustrative data or theoretical distribution. Exact plotted values remain editable.

Frequency polygon: comparison with common bins



How to read this graph

Both illustrative samples contain 20 observations with the same bins.

The second distribution is shifted toward higher concentration.

Identical class definitions are essential for the comparison.

Illustrative data or theoretical distribution. Exact plotted values remain editable.

Percentile interpolation | Try it

For ordered values 2,5,8,11,14,17, calculate the type-7 80th percentile.

Why might PERCENTILE.EXC give a different answer?

Percentile interpolation | Solution

$h = 1 + (6 - 1) \times .8 = 5$, so the answer is the fifth value, 14.

The exclusive rule uses a different plotting-position convention. Differences matter particularly for short samples and extreme percentiles.

R laboratory: reproducible analysis

```
x <- c(1,2,3,4,20)
quantile(x,c(.25,.5,.75),type=7)
boxplot(x,ylab="Nitrate (mg/L)")
plot(ecdf(x),xlab="Nitrate (mg/L)",ylab="Cumulative proportion")
```

Expected quartiles are 2, 3 and 4. Document the quartile rule used by a built-in boxplot.

Excel laboratory: formulas and interpretation

Put 1,2,3,4,20 in A2:A6. Use =QUARTILE.INC(A2:A6,1), then 2 and 3.

Create a scatterplot for paired variables. Do not use a line chart when x spacing is unequal.

Add axis titles and specify whether plotted uncertainty is SD, SE or a confidence interval.

Expected quartiles are 2, 3 and 4. Document the quartile rule used by a built-in boxplot.

Study references and further reading

OpenStax §§2.1–2.4; NIST EDA graphical techniques §1.3.3; EPA QA/G-9S Chapter 2

OpenStax: free introductory explanations and exercises. NIST: formulas, assumptions and graphical diagnostics.

EPA QA/G-9S: environmental data quality and method-selection context. Penn State STAT 414: probability derivations beyond introductory coverage.

Course-list texts: Spiegel for worked calculations; Hogg & Craig or Mood, Graybill & Boes for mathematical development. Editions vary, so use the relevant topic index rather than an unverified page number.

Open-access reading links

OpenStax: https://openstax.org/books/introductory-statistics-2e/pages/1-introduction

NIST: https://www.itl.nist.gov/div898/handbook/

EPA: https://www.epa.gov/quality/guidance-data-quality-assessment

Penn State: https://online.stat.psu.edu/stat414/

R: https://stat.ethz.ch/R-manual/R-devel/library/stats/html/00Index.html

Excel: https://support.microsoft.com/en-us/excel/statistical-functions-reference